Blush – Solution Approach & Design (Phase 1)

Salesforce Fresher Batch | Milestone 2: Solution Design | Prepared 23 Oct 2026 | Status: Draft for mentor review

1. Purpose and Scope

This document proposes how Phase 1 of the Blush Retail & Wholesale Management assignment will be built in Salesforce. It builds on the Milestone 1 document (Clarifying Questions and Assumptions, 21 Oct 2026). Question numbers Q1–Q31 refer to that document, so every design choice can be traced to an approved assumption. After mentor review this becomes the build baseline.

In scope: Requirements A–J of the assignment (customers and sales, campaigns, products and pricing, reminders, commission, order controls, invoice and GST, reports and dashboards, users and app, testing).

Out of scope (later phases): eBay integration, website ordering, partial payments, returns and input tax credit.

Item	Detail
Version	0.1 (draft)
Author	[Your name], Salesforce Fresher Batch
Reviewer	[Mentor name]
Org	Developer Edition, built with SFDX project in Git
Baseline	Milestone 1 assumptions Q1–Q31

How to read this: Section 2 gives the big picture, Section 3 the data model, Section 4 the solution for each requirement, Sections 5–7 the calculation rules, code design and security, Sections 8–9 reports and testing, and Section 10 the plan, risks and the decisions I need from you.

2. Solution Overview

2.1 Design principles

- **Standard first.** Lead, Account, Contact, Opportunity, Product, Price Book, Order, Campaign, Task, Event, approvals, reports and dashboards are used as they are. Custom objects are limited to Store__c and an error log.
- **Configuration over code, code where it is clearly needed.** Apex is used only where declarative tools cannot meet the rule (see 2.2). Each Apex piece has a stated reason.
- **Rules live in metadata, not in code.** GST slabs, commission thresholds, discount limit, home state, wholesale discount and the reminder window are stored in Custom Metadata, so a rule change is a data change, not a deployment.
- **Bulk-safe and idempotent.** Every trigger and job handles 200 records, uses no SOQL or DML inside loops, and can be re-run without creating duplicates.
- **Secure by default.** Least-privilege permission sets, private sharing where needed, field-level security on calculated fields, and Apex that respects sharing and FLS unless a documented exception applies.
- **Frozen financials.** Tax, totals and commission are stored on the Order and its lines, so a later price or rule change never alters a past invoice.

2.2 Declarative or Apex: the decision for each requirement

Need	Chosen tool	Reason
Record types, layouts, app, list views, permission sets	Declarative	Fully supported natively
Defaults: price book, store, Sales Rep, Order Owner	Before-save record-triggered Flows	Simple field defaults, fast, admin-maintainable
Create Order and copy products from Opportunity	Autolaunched Flow behind a quick action	Straight record copy with clear fault path

Need	Chosen tool	Reason
Customer matching and Lead conversion (Q9–Q12)	Apex trigger handler + service	Two-key priority matching, normalised phone, and Database.convertLead has no native Flow action; must be bulk-safe
Price book entry sync (Q16)	Apex trigger on Product2	Bulk upsert across Standard, Retail and Wholesale price books in the correct order
Line tax, totals, commission (Q4, Q5, Q31)	Apex calculator + trigger	Per-line half-up rounding, metadata-driven slabs; roll-up summary fields are not expected to be available on Order Products (to be verified in org)
Discount approval (Q18, Q19)	Approval Process + validation rule	Native approvals give locking and manager routing
Paid-order lock (Q8)	Validation rules + custom permission + Apex delete guard	Rules do not fire on delete, so a small guard is needed
Nightly reminders (Q20–Q22)	Batchable + Schedulable Apex	Needs Task and Event scan, flag and bulk create within governor limits
Invoice PDF (Q30)	Visualforce renderAs=pdf + controller extension	Native, no extra licence, conditional tax columns
Reports and dashboards	Declarative	Summary formulas, fiscal quarters and dashboard filters cover the needs

2.3 Logical architecture

```
Lead ──convert──> Account (Retail / Wholesale) ──< Contact
                      |
                      └──< Opportunity (Retail / Wholesale) >── Pricebook2
(Retail | Wholesale)
                              |  ├──< OpportunityLineItem >── PricebookEntry
>── Product2
                              |  ├──< Task / Event  (nightly reminder job)
                              |  └── 0..1 Order (Retail / Wholesale) >──
Store__c
                              |            └──< OrderItem (tax and net price
per line)
                              └── Owner = Sales Rep (copied to
Order.Sales_Rep__c)
Campaign ──< CampaignMember >── Lead / Contact
```

3. Data Model

3.1 Objects and record types

Object	Type	Record types	Notes
Lead	Standard	None (Q14)	Phone normalised for matching
Account	Standard	Retail (B2C), Wholesale (B2B)	Business Account model, Account Name = customer's full name (Q2)
Contact	Standard	None	One per customer, linked to Account
Opportunity	Standard	Retail, Wholesale	Captures name, email, phone
Product2, Pricebook2, PricebookEntry	Standard	None	Three price books: Standard, Retail, Wholesale
Order, OrderItem	Standard	Order: Retail, Wholesale	Orders feature enabled

Object	Type	Record types	Notes
Campaign	Standard	None	Marketing users only
Task, Event	Standard	None	Event gets a completion checkbox
Store__c	Custom	None	Store name, address, GSTIN (Q7)
Application_Log__c	Custom	None	Errors from triggers and batch jobs
Blush_Setting__mdt, GST_Rate__mdt	Custom metadata	n/a	Business rules as data

3.2 Key custom fields

Lead

Field	Type	Purpose
Phone_Normalized__c	Text(10)	Last 10 digits of phone, populated by trigger (Q11)
Possible_Duplicate_Of__c	Lookup(Lead)	Flags the second match when email and phone hit different leads (Q9)

Opportunity

Field	Type	Purpose
Customer_Name__c	Text(120)	Customer's full name (required)
Customer_Email__c	Email	Email, lower-cased on save
Customer_Phone__c	Phone	Phone as entered
Phone_Normalized__c	Text(10)	System field for matching
Customer_Link_Status__c	Picklist	Pending, Linked, Failed (makes processing idempotent and retryable)
Customer_Link_Message__c	Text(255)	Reason shown to the rep on failure

Field	Type	Purpose
Matched_Lead__c	Lookup(Lead)	Lead that was converted, for audit
Follow_Up_Reminder_Created__c	Checkbox	One reminder ever per opportunity (Q21)

Validation: Customer name is mandatory, and at least one of email or phone is mandatory.

Product2

Field	Type	Purpose
Product_Key__c	Text, External ID, Unique	Loaded from Product Code (for example 10#025) so loads are repeatable
Category__c	Picklist	Garments or Jewellery (Jwellery corrected, per Milestone 1)
Available_Retail__c, Available_Wholesale__c	Checkbox	Channel availability
Retail_Price__c	Currency	Base price from the product file
Wholesale_Price__c	Currency	Retail price less 30%, set by trigger from metadata, read-only
Cost_Price__c	Currency	Cost, visible to managers and finance only

Pricebook2: Channel__c (Picklist: Retail, Wholesale) so rules compare channel, not price book names.

Order

Field	Type	Purpose
Sales_Rep__c	Lookup(User)	Fixed copy of the Opportunity Owner (Q1)
Store__c	Lookup(Store__c)	Selling store (Q7)

Field	Type	Purpose
Payment_Mode__c	Picklist	Cash, UPI, Card, Bank Transfer
Invoice_Number__c	Auto Number INV-{0000}	Invoice number (Q30)
Invoice_Date__c	Date	Set when the order is first activated
Approval_Status__c	Picklist	Not Required, Pending, Approved, Rejected
Max_Line_Discount_Percent__c	Percent	Highest line discount, drives approval (Q18)
Intra_State__c	Formula (Checkbox)	True when Billing State equals the home state in metadata
Effective_Order_Amount__c	Currency	Sum of line taxable amounts, excluding GST (Q5, Q6)
Total_CGST__c, Total_SGST__c, Total_IGST__c, Total_GST__c	Currency	GST totals
Grand_Total__c	Currency	Total amount payable
Commission_Amount__c	Currency	Frozen once the order is Paid
Paid_Date__c	Date	Set when Status becomes Paid

OrderItem: Discount_Percent__c, Product_Category__c (snapshot), Discounted_Unit_Price__c, Taxable_Amount__c, GST_Rate__c, CGST_Amount__c, SGST_Amount__c, IGST_Amount__c, Tax_Amount__c, Net_Price__c, Line_Cost__c (cost snapshot for revenue and profit reporting).

Other: Event.Is_Completed__c (Checkbox), Task.Is_Follow_Up_Reminder__c (Checkbox, so the job ignores its own tasks), User.Default_Store__c (Lookup), Campaign.Active_This_Month__c (Formula Checkbox).

Store__c: Name, Street, City, State, Postal Code, GSTIN__c (15 characters, format validation), Phone__c, Is_Active__c.

Custom metadata: Blush_Setting__mdt holds the discount limit (10), wholesale discount (30), home state (Maharashtra), commission threshold (1000), commission rate (1%), flat commission (30) and reminder window (5 days). GST_Rate__mdt holds rows of Category, Threshold and Rate: Garments 0 → 5%, Garments 1000 → 12%, Jewellery 0 → 3%. The rule picks the row with the highest threshold that the discounted unit price is strictly above.

3.3 Standard field and platform choices

- State and Country picklists are enabled with Indian states, so “Maharashtra” is never a free-text mismatch that breaks CGST/SGST logic.
- Field history tracking is turned on for Order Status, Approval Status, Sales Rep and discounts, for audit.
- Orders are enabled in Setup. Order Status values: Draft (processing), Activated (confirmed), Paid, Cancelled.

4. Solution Approach by Requirement

4.1 Requirement A: Customer and Sales Management

Approach. Lead, Account, Contact, Opportunity and Order are configured as in the assignment table. Retail and Wholesale are separated by record type on Account, Opportunity and Order, with its own page layout, compact layout and list views, so the two are always visible and reportable as different.

Opportunity capture. Custom fields on Opportunity hold name, email and phone, because Opportunity has no native fields for them and the customer may not exist yet (Q13).

Processing logic (Apex `OpportunityTrigger` → `CustomerResolutionService`, runs after insert when status is Pending):

1. Normalise: email trimmed and lower-cased, phone reduced to its last 10 digits (Q11).
2. Match Leads in bulk with two queries: email first, then phone, unconverted leads only. If email and phone match different leads, convert the email match and set Possible_Duplicate_Of__c on the other (Q9).

3. If a lead matches, convert it with `Database.convertLead` with no new Opportunity created. If an Account or Contact already exists with that email or phone and the same record type, convert into it (Q10).
4. If no lead matches, reuse a matching Account and Contact of the same record type, otherwise create an Account (record type from the Opportunity, Name = full name) and a Contact.
5. After conversion, set the Account record type to the Opportunity's (convertLead uses the user's default) (Q12), set Opportunity.AccountId, create the primary Opportunity Contact Role, store Matched_Lead__c and set status Linked.
6. On any failure, the Opportunity still saves; status becomes Failed with a readable message and an entry goes to Application_Log__c. Setting the status back to Pending retries.

Guard rails. Standard duplicate and matching rules on Lead, Account and Contact warn on manual entry. The matching service queries without sharing so it can see leads owned by other reps (it returns only record Ids and is documented as a deliberate exception). Leads over 100 in one transaction are chunked and the rest run in a Queueable job.

Order creation. A quick action "Create Order" on Opportunity runs an autolaunched Flow that creates the Order (record type matching the Opportunity, Draft status, same price book, Account, billing address from Account, Store from the user's default, Sales_Rep__c from the Opportunity Owner) and copies the line items. One Opportunity has at most one Order in Phase 1.

4.2 Requirement B: Promotions and Campaigns

Approach. Standard Campaign object. Users with the Marketing User checkbox and the Marketing permission set can create and edit campaigns. A formula checkbox `Active_This_Month__c` is true when the campaign starts on or before the end of this month and ends on or after the start of this month, or has no end date (Q28). A list view "Active This Month" and a report use this field, and it appears on the Marketing dashboard. Sales users get read-only access to campaigns.

4.3 Requirement C: Product Catalogue and Pricing

Approach. Product2 with the custom fields in 3.2. The ten valid products from Products.xlsx are loaded by Data Loader using Product_Key__c as the External ID. The blank last row is excluded, and all ten products are available to both channels (Milestone 1). Wholesale price is 70% of base price, which matches the supplied sheet.

Pricing structure. Standard price book plus Retail and Wholesale price books, identified by Pricebook2.Channel__c.

Auto-setup for new products (Q15, Q16). Apex `Product2Trigger` → `PricebookSyncService`, bulk and idempotent:

- When a product is saved, Wholesale_Price__c is recalculated from metadata.
- If either channel is ticked, a Standard price book entry (retail price) is ensured first, because the platform requires it before any custom price book entry.
- A Retail entry (retail price) and a Wholesale entry (wholesale price) are inserted or updated according to the ticked channels. Unticking a channel or deactivating the product marks the entry inactive, never deletes it, so historical order lines stay valid.
- When the base price changes, existing entries are updated. Lines already on opportunities and orders keep their old price.

Channel enforcement. A before-save Flow sets the Opportunity price book from its record type. A validation rule blocks a mismatch between Opportunity record type, Pricebook2.Channel__c and Account record type. A rep can only add products that have an active entry in that price book, which is native behaviour. This delivers “retail opportunities use retail classification and retail pricing” without any admin action.

4.4 Requirement D: Follow-up Reminders

Approach. `FollowUpReminderBatch` (Database.Batchable) with `FollowUpReminderScheduler` (Schedulable), scheduled nightly at 01:00 through the Scheduled Jobs page.

- **Start query:** open Opportunities (IsClosed = false) where Follow_Up_Reminder_Created__c is false.
- **Execute (200 per batch):** one query for incomplete Tasks (IsClosed = false) and one for Events (Is_Completed__c = false) on those Opportunities, dated from today minus 5 days through today inclusive. Future dates are ignored (Q20). Tasks flagged Is_Follow_Up_Reminder__c are excluded so the job never triggers itself.
- **Action:** for each qualifying opportunity create one Task: Subject “Follow-up: <Opportunity Name>”, due today, High priority, Not Started, owned by the Opportunity Owner (Q22). Then set Follow_Up_Reminder_Created__c, so each opportunity gets one reminder ever (Q21).
- **Separate queries are used for Tasks and Events** because a SOQL semi-join cannot be combined with OR.

- **Failure handling:** `Database.insert(records, false)`; the flag is set only for opportunities whose Task was created; failures go to Application_Log__c and are retried next night.
- **Example check:** on 15 July an activity dated 12 July qualifies; one dated 1 July does not.

4.5 Requirement E: Order Commission

Approach. `OrderCalculationService` calculates per order, not per line (Q5). Effective Order Amount is the sum of line taxable amounts after discount and before GST (Q6). Commission is 1% of that amount if it is 1,000 or less, otherwise a flat 30, rounded half-up to 2 decimals. The thresholds come from metadata. Commission is recalculated while the order is not Paid and frozen once it is Paid. Cancelled orders are excluded from commission reporting (Q27). The commission rep is Order.Sales_Rep__c (Q1).

4.6 Requirement F: Order Controls

Approach. Sales users can edit orders and order lines while the status is Draft. When Status becomes Paid:

- A validation rule on Order blocks any change when the prior status was Paid, unless the user has the custom permission `Edit_Paid_Orders`.
- A validation rule on OrderItem blocks insert and update when the parent Order is Paid, with the same exception.
- An `OrderItemTrigger` before-delete guard blocks deletion of lines on Paid orders, because validation rules do not run on delete.
- Paid requires Payment Mode, Store and a billing state, and sets Paid_Date__c. Paid cannot be reversed except by users with the custom permission, which is given to a finance/admin permission set (Q8).

A platform behaviour to verify early: standard Orders restrict changes to products once an Order is in the Activated category. I will test this in the org in the first build week. If it blocks normal sales work, the fallback is to keep the standard Status in Draft and drive confirmation with a custom Payment Status field.

4.7 Requirement G: Invoice, Discount and Tax

Discount and approval (Q18, Q19). Discount is a percentage per line (Discount_Percent__c, 0 to 100). The trigger stores the highest line discount on the Order. If any line exceeds 10%, Approval_Status__c becomes Pending-required; an Approval Process on Order (entry criteria: Max_Line_Discount_Percent__c greater than 10) sends it

to the submitter's manager. The record is locked while pending. A validation rule prevents moving to Activated or Paid until approval is Approved. Rejection returns the order to Draft with the reason.

Tax (Q3, Q4, Q31). Prices are tax-exclusive. For each line: discounted unit price, taxable amount, GST rate chosen by category and slab on the discounted unit price, and tax rounded half-up to 2 decimals. If the Order's billing state is the home state, CGST is half the tax rounded half-up and SGST is the remainder, otherwise IGST is the whole tax. Net price is the taxable amount plus tax, which equals (Unit Price − Discount) × Quantity + Taxes. Worked examples are in Section 5.

Invoice PDF. A Visualforce page `OrderInvoice` with `renderAs="pdf"` and a controller extension, opened from a "Generate Invoice" button on the Order. It follows Invoice_Format.xlsx with the extensions agreed in Milestone 1: Quantity, IGST, Store and Payment Mode.

- Header: Blush logo, store name, address, GSTIN and phone from the Store record (the template's blanks come from Store__c).
- Invoice number and date, customer name and phone from the Account.
- Line table with serial number, product code, description, quantity, unit price, discount, tax columns, and net price. CGST and SGST columns show only for intra-state orders, and IGST only for inter-state orders.
- Total amount payable, payment mode, and the two footer notes from the template.
- The button is available only when the order is Activated or Paid, and the page shows a clear message if Store, Payment Mode or billing state is missing.
- The PDF can be saved to the Order as a File, so the exact invoice sent is retained.